Two follow-up projects: the binary-measurement majorization protocol on quantum light sources

Technical note for two Master's research projects

Prepared for M. F. Ciappina

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Abstract

We propose two Master's research projects that take the majorization uncertainty protocol of Zhu, Gour and Ciappina [1] for two-outcome measurements from qubits on paper to real quantum light. *Project A* (discrete variables) tests the draft's biased-POVM and steering results with heralded single photons and polarization-entangled pairs read out by threshold detectors. The student derives the optimal direct-sum vector of the measurement actually performed, including dark counts, click-assignment rules and multi-photon inputs; measures Lorenz curves and the non-attainment gap; maps where mismatched detectors break the ideal relation; and runs a steering test that first exhibits and then removes a spurious violation. *Project B* (continuous variables) answers open question (v) of the draft with a positive example. Periodically binned homodyne outcomes are binary measurements on a light mode with $c < 1$. When the two bin periods satisfy $T_x T_p = 4\pi$ ($\hbar = 1$), the binned quadratures act as Pauli Z and X on a GKP-type subsystem qubit, so $c = 1/2$ exactly and every qubit result of the draft transfers. The student maps c over the period ratio, models homodyne loss (unsharpness) and duty-cycle errors (bias), and applies the bounds to squeezed vacuum, to cat and GKP-type states and to EPR steering with two-mode squeezed light. Both projects have a numerical core that stands on its own; preliminary numerical checks behind every number quoted here are provided with this note.

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1 Background: the protocol in the draft

The protocol of Ref. [1] turns any pair of two-outcome measurements M and N into one state-independent probability vector ω that majorizes the outcome statistics of every state,

$$p(\rho) \oplus q(\rho) \prec \omega_{\oplus} \quad \text{or} \quad p(\rho) \otimes q(\rho) \prec \omega_{\otimes} \qquad \forall \rho. \tag{1}$$

One calibration therefore bounds the Shannon, Rényi, Tsallis and every other Schur-concave uncertainty measure at once. Both projects reuse the same five steps.

Step 1. Characterize the measurements. A projective qubit pair is fixed by its overlap $c = \cos^2(\theta/2)$, with θ the Bloch angle between the axes. A general two-outcome qubit POVM is

$$E_0 = \tfrac{1}{2}\big[(1+\mu)\mathbb{1} + \eta\,\hat{a}\cdot\vec{\sigma}\big], \qquad \eta \geq 0, \quad |\mu| + \eta \leq 1, \tag{2}$$

with sharpness η [16] and bias μ; the second POVM has parameters $(\nu, \zeta, \hat{b})$.

Step 2. Compute the maximal partial sums $S_k = \max_\rho L_k$ and their increments. For projective qubits (Theorems 1 and 2 of Ref. [1]; see also Refs. [2, 4]),

$$\omega_\oplus = (1,\ \sqrt{c},\ 1-\sqrt{c},\ 0), \qquad \omega_\otimes = (w,\ 1-w,\ 0,\ 0), \qquad w = \tfrac{1}{4}\big(1+\sqrt{c}\big)^2. \tag{3}$$

For noisy and biased POVMs (Theorem 4),

$$\omega_\oplus(M,N) = \tfrac{1}{2}\big(1+\kappa,\ 1+\Lambda-\kappa,\ 1+\kappa-\Lambda,\ 1-\kappa\big), \tag{4}$$

$$\kappa = \max\{|\mu|+\eta,\ |\nu|+\zeta\}, \qquad \Lambda = \max_{s,t=\pm1}\big[s\mu + t\nu + |s\eta\hat{a} + t\zeta\hat{b}|\big]. \tag{5}$$

In any dimension (Corollary 4, building on the eigenvalue formula of Ref. [3]), S_1 is the largest eigenvalue among the four effects and $S_2 = \max_{i,j}\lambda_{\max}(E_i + F_j)$, which gives the optimal direct-sum vector numerically.

Step 3. Check attainment. Unless the effects commute, no state reaches ω (Propositions 1 and 3), so every induced bound for a continuous strictly Schur-concave measure is strict (Theorem 3). The distance

$$\Delta = \min_\rho \max_k \big[S_k - L_k\big(p(\rho)\oplus q(\rho)\big)\big] \tag{6}$$

between the bound and the closest state is measurable in principle.

Step 4. Check the detectors. Unbiased noise never breaks an ideal-detector bound. Bias does: the ideal relation $p \oplus q \prec \omega_\oplus$ survives if and only if $\Lambda \leq 2\sqrt{c}$ (Proposition 4), and click/no-click detection breaks it for every efficiency $\eta_d < 1$.

Step 5. Use it as a witness. Every local-hidden-state model obeys (Corollary 3)

$$H(M|A_1) + H(N|A_2) \ \geq\ \min_\rho\big[H(M)+H(N)\big] \ \geq\ H\big(\omega_\oplus(M,N)\big). \tag{7}$$

For a Werner state with click/no-click detection at Bob, the majorization witness needs $\eta_d > \eta^* = (4+\sqrt{2})/7 \approx 0.7735$ and the exact Shannon witness needs $\eta_d > \eta_S \approx 0.6806$.

Why quantum light. Single-photon polarization realizes the qubit case exactly: two linear analyzers at relative angle ϑ give $\theta = 2\vartheta$ and $c = \cos^2\vartheta$ (Malus's law). Real photon counters are exactly the biased, unsharp POVMs of Theorem 4; a polarizing beam splitter (PBS) followed by detectors of efficiencies η_H, η_V, with no-click events split evenly, has $\mu = (\eta_H - \eta_V)/2$ and $\eta = (\eta_H + \eta_V)/2$, so the efficiency mismatch known from quantum key distribution [7] appears as a bias. Earlier experiments with coherent light [5] and with photons [6] tested ideal-detector bounds only, not the biased-POVM and steering tests introduced in Ref. [1]. The draft also leaves open which binary measurements on a light mode have $c < 1$, because parity, displaced parity and on/off detection after a displacement all have $c = 1$. Project A addresses the first gap and Project B the second.

2 Project A: detector-aware majorization and steering tests with single photons

Project A delivers the first test of the draft's biased-POVM results on real photons: the optimal bound for the polarization measurement actually performed, measured Lorenz curves with the non-attainment gap, a map of where mismatched detectors break the ideal relation, and a steering test that shows and then removes a spurious violation.

2.1 Aims

A1. Derive the optimal direct-sum vector for a polarization analyzer built from a PBS and two threshold detectors, including efficiencies, dark counts, no-click and double-click assignment rules, and multi-photon inputs.

A2. Measure Lorenz curves of $p \oplus q$ for heralded single photons and show that no prepared state reaches $\omega_\oplus$ (Proposition 1), by measuring the gap Δ of Eq. (6).

A3. Scan deliberately mismatched detector efficiencies and map where the ideal relation fails (Proposition 4), including the same-basis trade-off of Proposition 3(iii).

A4. Run a one-sided steering test with polarization-entangled pairs, Alice untrusted and Bob trusted with click/no-click detection. Show the spurious violation of the ideal bound by a product state, its removal by the corrected bound, and the Werner-state thresholds η^* and η_S.

2.2 Protocol adaptation

Single-photon detector model. For one photon the analyzer is a two-outcome qubit POVM, so Theorem 4 applies directly. Let η_H, η_V be the detector efficiencies, d_H, d_V the dark-count probabilities per coincidence window, no-click events split 50/50, and double clicks assigned to outcome H with probability q. The effect of outcome H is diagonal with entries

$$a = c_H(1-d_V) + q\,c_H d_V + \tfrac{1}{2}(1-c_H)(1-d_V), \qquad c_H = 1-(1-d_H)(1-\eta_H), \tag{8}$$

$$b = d_H(1-c_V) + q\,d_H c_V + \tfrac{1}{2}(1-d_H)(1-c_V), \qquad c_V = 1-(1-d_V)(1-\eta_V), \tag{9}$$

so that $\mu = a + b - 1$ and $\eta = a - b$. Without dark counts this reduces to $\mu = (\eta_H - \eta_V)/2$ and $\eta = (\eta_H + \eta_V)/2$, as in Ref. [1]. Rotating the analyzer with a half-wave plate gives the second POVM, and κ, Λ and $\omega_\oplus(M, N)$ follow in closed form from Eq. (4).

Multi-photon inputs. An SPDC source also emits two or more photons, so Bob's input lives in the two-mode Fock space. The click statistics are still a binary POVM, block-diagonal in the total photon number n. In the block n, with basis $|n-k, k\rangle$ ($n_H = n-k$, $n_V = k$),

$$E_H^{(n)} = \sum_{k=0}^{n} P(H \,|\, n-k, k)\,|n-k, k\rangle\langle n-k, k|, \qquad c_H(n_H) = 1-(1-d_H)(1-\eta_H)^{n_H}, \tag{10}$$

and the rotated setting is $F_H^{(n)} = R_n(\theta) E_H^{(n)} R_n(\theta)^\dagger$, with $R_n(\theta) = e^{-i\theta J_y}$ in the spin-$n/2$ representation. Corollary 4 then gives

$$S_1 = \max_n \max_X \lambda_{\max}(X^{(n)}), \qquad S_2 = \max_n \max_{i,j} \lambda_{\max}(E_i^{(n)} + F_j^{(n)}). \tag{11}$$

A first numerical check (photon numbers up to 30, converged by $n = 10$) gives Table 1.

Three preliminary conclusions follow, to be confirmed and extended by the student. (i) With equal efficiencies and random double-click assignment, the ideal relation survives in the full Fock

Table 1: Second maximal partial sum S_2 for a PBS with two threshold detectors of equal efficiency and random double-click assignment. The ideal relation survives if $S_2 \leq 1 + \sqrt{c}$.

Bloch angle θ	Efficiency	Ideal limit $1 + \sqrt{c}$	S_2, one photon (Thm. 4)	S_2, Fock (Cor. 4)
90°	0.50	1.7071	1.3536	1.6324
90°	0.80	1.7071	1.5657	1.6788
90°	0.95	1.7071	1.6718	1.7053
60°	0.50	1.8660	1.4330	1.8022
60°	0.80	1.8660	1.6928	1.8336
60°	0.95	1.8660	1.8227	1.8639

space. (ii) The tighter single-photon vector of Theorem 4 does not: multi-photon inputs push S_1 to 1 and S_2 well above its single-photon value, so a test that uses the single-photon corrected bound on an SPDC source can be fooled. (iii) Assigning double clicks deterministically gives $S_2 = 2$ and breaks every nontrivial bound, and an efficiency mismatch of 0.9 against 0.7 breaks the ideal relation even with random assignment ($S_2 = 1.753$ at $\theta = 90°$). This is the majorization counterpart of the squashing question for optical detectors [8].

Steering. Alice measures σ_z and σ_x with her own detectors and heralds Bob's photon; her side is untrusted, so her losses do not matter. Bob's no-click events are kept, which removes the fair-sampling assumption on his side. Eq. (7), with $\omega_{\oplus}(M, N)$ of Bob's real POVM computed in Fock space, is the witness [9, 10, 11]; photonic steering tests of this one-sided kind are well established [12].

2.3 Methods

- **Theory and numerics.** Closed forms from Theorem 4; eigenvalue problems from Corollary 4 in a truncated Fock space (dimension 66 at $n \leq 10$); Lorenz curves and the lattice supremum via least concave majorants; exact entropy minima on the great circle refined by Nelder–Mead, as in Appendix A of Ref. [1]. Open question (i) of the draft, the tensor-product vector for noisy POVMs, is an optional numerical extension.
- **Source.** A type-II ppKTP crystal in a Sagnac loop pumped at 405 nm gives polarization-entangled pairs at 810 nm [13]; heralding on one arm gives single photons. A quantum-dot source is an alternative if multi-photon contamination must be minimal.
- **State preparation and analysis.** Polarizer, half- and quarter-wave plates on motorized mounts prepare states on the great circle. The analyzer is a half-wave plate, a PBS and two fiber-coupled detectors (SPADs near 0.6 efficiency, or SNSPDs near 0.9) read by a time tagger with a coincidence window of about 1 ns.
- **Bias on demand.** A calibrated variable fiber attenuator in one PBS output sets the mismatch; blocking the V output gives click/no-click detection. Efficiencies are calibrated by the Klyshko method [15] and dark counts by blocked-input runs.
- **Werner states.** Mixing data sets taken with $|\Phi^+\rangle$ and with the four product states $|HH\rangle, |HV\rangle, |VH\rangle, |VV\rangle$ in the right proportions realizes $w|\Phi^+\rangle\langle\Phi^+| + (1-w)\mathbb{1}/4$ exactly.
- **Statistics.** Maximum-likelihood state reconstruction [14], bootstrap confidence intervals, and a systematic-error budget for wave-plate angles (about 0.1°) and efficiency calibration (about 1%).

2.4 Numerical and experimental work packages

WP-A1. Reproduce Figs. 2, 5 and 6 of Ref. [1]; build the detector-model code with dark counts and assignment rules.

WP-A2. Fock-space analysis with Corollary 4: multi-pair emission at a given mean pair number, setting-dependent loss, and a "safe bound" calculator for any measured detector calibration.

WP-A3. Single-photon runs: Lorenz curves at analyzer angles $\vartheta = 22.5°, 30°, 33.5°, 37.5°, 45°$; the gap Δ; an efficiency-mismatch scan for Proposition 4; and the same-basis trade-off of Proposition 3(iii).

WP-A4. Steering runs: a product state with Bob in the dark eigenstate, the Bell state, and Werner states, with Bob's attenuation scanned through η_S and η^*.

WP-A5. Thesis and a paper draft.

If no lab is available, WP-A3 and WP-A4 run as a virtual experiment: Monte Carlo counts with calibrated source and detector parameters, analyzed by the same pipeline. The thesis then hands a partner lab a ready analysis chain.

2.5 Expected results

The gap between $\omega_\oplus$ and the closest state is small, so it sets the statistics needed. Minimizing over states on the great circle (where the minimum lies, by the radial argument of Ref. [1]) gives Table 2.

Table 2: Non-attainment gap Δ for projective qubit measurements and the approximate number of detected photons per setting needed to resolve it at 5σ from counting statistics alone.

Bloch angle θ	Analyzer angle $\vartheta = \theta/2$	Overlap c	Gap Δ	Photons per setting
90°	45°	0.500	0.045	6×10^3
75°	37.5°	0.629	0.033	1×10^4
67.1°	33.5°	0.695	0.027	2×10^4
60°	30°	0.750	0.022	3×10^4
45°	22.5°	0.854	0.013	7×10^4

At heralded rates of 10^4 to 10^5 per second, statistics take seconds to minutes per setting, so systematic errors, not counts, will limit the gap measurement; $\theta = 90°$ is the safest first target.

- **Proposition 4 map.** A measured version of Fig. 6(a) of Ref. [1]. For $\eta_H = 0.95$, the ideal relation should fail below $\eta_V \approx 0.708$ and the ideal one-bit Shannon bound below $\eta_V \approx 0.211$, while equal efficiencies never violate either.
- **Spurious steering.** A product state with Bob in the dark eigenstate gives $H(M|A_1) + H(N|A_2) = h(\eta_d/2)$: 0.881 bit at $\eta_d = 0.6$ and 0.993 bit at $\eta_d = 0.9$. Both fall below the ideal one-bit bound and would be reported as steering; the corrected bound removes the violation.
- **Werner-state thresholds.** The Bell-state witness should switch on at $\eta_d \approx 0.681$ for the exact Shannon criterion (our recomputation gives 0.6806, matching Ref. [1]) and at $\eta_d \approx 0.773$ for the majorization criterion.
- **Multi-photon correction.** A quantified penalty, in bits, for using the single-photon bound at a given mean pair number, and the Fock-space bound that removes it.

2.6 Risks and mitigations

- **Systematics larger than Δ.** Start at $\theta = 90°$, randomize the order of settings, and propagate the calibration uncertainty into S_k.
- **Multi-pair emission.** Keep the mean pair number per window near 0.01 and always report the Fock-space bound.
- **Efficiency drift between settings.** Interleave calibration runs; the draft flags drift and setting-dependent loss as needing an extended model, which WP-A2 provides.
- **No lab access.** The virtual experiment keeps every theoretical deliverable intact.

3 Project B: majorization uncertainty for binned quadratures of squeezed and EPR light

Project B answers open question (v) of Ref. [1] with a positive example: periodically binned homodyne outcomes are binary measurements on a light mode with $c < 1$, and at a period product of 4π they reproduce the qubit theory exactly. The student then applies the bounds to squeezed vacuum, to non-Gaussian states (cat and GKP-type states) and to EPR steering with two-mode squeezed light.

3.1 Aims

B1. Classify accessible binary measurements on a light mode by their overlap c: parity, displaced parity and on/off detection after a displacement (all $c = 1$, as the draft shows) and pseudospin operators [29], against periodically binned quadratures.

B2. Compute c, $\omega_\oplus$ and $\omega_\otimes$ for binned $\hat{x}$ and $\hat{p}$ as a function of the period product, establish attainment, and identify the states that saturate each partial sum.

B3. Model real homodyne detection: finite efficiency (unsharpness), bin duty-cycle errors (bias), bin offsets, and finite energy.

B4. Apply the bounds to single-mode squeezed vacuum, to cat and GKP-type states, and to EPR steering with two-mode squeezed vacuum, and compare with the Reid criterion [25] and continuous entropic criteria [9].

3.2 Protocol adaptation

The binary measurement. With $\hbar = 1$, $[\hat{x}, \hat{p}] = i$ and vacuum variance $1/2$, define the projectors

$$P_0 = \chi(\hat{x} \bmod T_x \in [0, T_x/2)), \qquad Q_0 = \chi(\hat{p} \bmod T_p \in [0, T_p/2)), \tag{12}$$

and $P_1 = \mathbb{1} - P_0$, $Q_1 = \mathbb{1} - Q_0$. Both are functions of the Weyl operators $U = e^{2\pi i \hat{x}/T_x}$ and $V = e^{2\pi i \hat{p}/T_p}$, which obey

$$UV = e^{-2\pi i r}\, VU, \qquad r = \frac{2\pi}{T_x T_p}. \tag{13}$$

For rational $r = p'/q'$ in lowest terms, the Zak transform [20, 19] splits the mode into q'-dimensional blocks labeled by two quasi-momenta (α, β), on which $U \simeq e^{i\alpha} C$ and $V \simeq e^{i\beta} S$ with C and S the clock and shift matrices. Theorem 5 of Ref. [1] then applies with

$$c = \sup_{\alpha,\beta} \max_{i,j} \left\| P_i Q_j \right\|^2. \tag{14}$$

The case $T_xT_p = 4\pi$. At $r = 1/2$, for example $T_x = T_p = 2\sqrt{\pi} \approx 3.545$, U and V anticommute. Every irreducible block is a qubit with $U \simeq e^{i\alpha}\sigma_z$ and $V \simeq e^{i\beta}\sigma_x$. Because exactly one of $\pm e^{i\alpha}$ lies in the upper half circle, P_0 restricted to a block is $|0\rangle\langle 0|$ or $|1\rangle\langle 1|$, and likewise Q_0 is $|+\rangle\langle +|$ or $|-\rangle\langle -|$. The two binnings are therefore exactly Pauli Z and X measurements on the logical qubit of the GKP code [21], i.e. of the modular subsystem decomposition of Ref. [22]. Hence $c = 1/2$, and every qubit result of Ref. [1] at $\theta = 90^\circ$ transfers:

$$p \oplus q \prec \left(1, \tfrac{1}{\sqrt{2}}, 1 - \tfrac{1}{\sqrt{2}}, 0\right), \qquad H(X_{\text{bin}}) + H(P_{\text{bin}}) \geq 1 \text{ bit}, \tag{15}$$

together with strictness and the two families of saturating states.

First numerical results. Computing c block by block for rational r with denominators up to 12 gives a discontinuous function (Table 3). Every odd denominator gives $c = 1$: in a block of odd dimension one projector of each pair has rank $(q' + 1)/2$, two such ranges must intersect, and the common eigenvector forces $c = 1$. Even denominators give $c < 1$, and only $r = 1/2$ reaches $c = 1/2$.

Table 3: Overlap of periodically binned quadratures for rational $r = 2\pi/(T_xT_p)$ (both duty cycles 1/2). The Shannon bound for $r \neq 1/2$ assumes the Jordan-block reduction. The last column is the two-mode squeezing needed for an EPR-steering violation with ideal detectors, Alice measuring continuous outcomes and Bob binning with $T_x = T_p$.

r	T_xT_p	Overlap c	Exact Shannon bound (bits)	Steering threshold
1/2	4π	0.500	1.000	9.3 dB
1/4	8π	0.854	0.467	13.0 dB
3/8	$16\pi/3$	0.897	0.353	17.2 dB
5/12	$24\pi/5$	0.914	–	–
3/10	$20\pi/3$	0.947	–	–
1/6	12π	0.971	0.123	22.8 dB
odd q'	$2\pi q'/p'$	1	0 (trivial)	none

For irrational r, and for any physical state of finite energy, the relevant quantity is an energy-constrained overlap $c_E(r)$, computed over states with mean photon number at most $\bar{n}$. Computing it is the main open theoretical question of the project.

Explicit-state checks at $r = 1/2$. Three thousand random states with up to 40 photons never cross the Lorenz curve of $\omega_\oplus$; their smallest Shannon sum is 1.44 bits. GKP-type grid states with peak width 0.3 (mean photon number about 5.5) reach $H(X_{\text{bin}}) + H(P_{\text{bin}}) = 1.0005$ bits, and their superposition at the bisector of the logical Bloch sphere (the GKP magic state) reaches $L_2 = 1.7070$ against the supremum $1 + 1/\sqrt{2} = 1.7071$. No finite-energy state reaches either bound, as Theorem 3 of Ref. [1] requires. The dictionary is thus: eigenstates of M or N $\leftrightarrow$ GKP logical states; bisector states $\leftrightarrow$ GKP magic states.

A binary-readout non-Gaussianity witness. Over all Gaussian states, the largest $L_2 = \max_{i,j}(p_i + q_j)$ we find is 1.580, reached by a coherent state centered in a bin. Because L_2 is convex in the state, mixtures of Gaussian states cannot exceed it either. Any

$$L_2 > L_2^{\text{G}} \approx 1.580 \tag{16}$$

therefore certifies a non-Gaussian state from binary data alone: an even cat state with $\alpha \approx 1.1$ gives 1.649, an odd cat with $\alpha \approx 2.4$ gives 1.620, and GKP-type states approach 1.707. This links directly to small optical cat states [30], including those generated in intense laser–matter interactions [31]. Proving the Gaussian bound analytically is part of aim B2.

Homodyne efficiency is unsharpness. With efficiency η, the rescaled readout is $\hat{x} + \xi$ with Gaussian noise of variance $\sigma^2 = (1-\eta)/(2\eta)$. The effect of outcome 0 becomes $E_0 = g(\hat{x})$, with g the square wave smoothed by the Gaussian. Since $g(x) + g(x + T/2) = 1$, every qubit block sees an *unbiased* unsharp Pauli measurement whose sharpness $|2g - 1|$ depends on the quasi-momentum. The optimal vector is Eq. (4) with the largest sharpness, which for $T = 2\sqrt{\pi}$ is

$$s_{\max}(\eta) = \frac{4}{\pi} \sum_{k \text{ odd}} \frac{(-1)^{(k-1)/2}}{k} e^{-\pi k^2 \sigma^2/2} = \begin{cases} 1.000 & \eta = 0.95 \\ 0.9997 & \eta = 0.90 \\ 0.976 & \eta = 0.80 \\ 0.889 & \eta = 0.70 \end{cases} . \tag{17}$$

The ideal relation survives, which is the analogue of Proposition 4(i).

Duty-cycle errors are bias. If bin 0 has width $(1/2 + \delta)T$, each block sees a biased POVM. With perfectly sharp binning, any $\delta \neq 0$ creates blocks in which one measurement is trivial, so $c = 1$ and the ideal relation fails at once. Homodyne loss softens this: the ideal relation survives ($\Lambda \leq \sqrt{2}$ in every block) up to $\delta \approx 1.4\%$ of the period at $\eta = 0.95$, 2.9% at 0.90 and 6.2% at 0.80. Better detectors make the ideal bound more fragile to binning bias. A shift of the bin boundaries is harmless, since it is a displacement and leaves c unchanged.

EPR steering. Alice homodynes with continuous outcomes (untrusted), Bob bins with $T_x = T_p = 2\sqrt{\pi}$ (trusted), and Eq. (7) is the witness. Treating Bob's efficiency as part of his calibrated measurement gives Table 4.

Table 4: Two-mode squeezing needed to certify steering of a two-mode squeezed vacuum with binned detection at Bob. "Corrected" uses the exact Shannon minimum (in bits) of the unsharp measurement actually performed; the last two columns give the squeezing needed.

Bob's efficiency	$s_{\max}$	Corrected bound	Beat ideal bound	Beat corrected bound
1.00	1.000	1.000	9.3 dB	9.3 dB
0.95	1.000	1.000	10.4 dB	10.4 dB
0.90	0.9997	1.002	12.1 dB	12.1 dB
0.80	0.976	1.095	not reached	17.9 dB
0.70	0.889	1.310	not reached	not reached

The corrected bound is the message of Ref. [1] in practice: at 80% efficiency only the bound of the measurement actually performed certifies steering. The binned witness is weaker than the Reid criterion [25], which detects any nonzero squeezing for a pure two-mode squeezed state. Its value is one bit per shot at Bob and a bound for every Schur-concave measure at once; the student should quantify whether Rényi or Tsallis versions lower the threshold. Squeezing of 15 dB [27] and two-mode squeezing of 10 dB [28] have been reported, which brackets these thresholds.

3.3 Methods

- **Theory.** Weyl operators, Zak transform and Jordan's lemma [17, 18] for the block structure; Theorems 4 and 5 and Corollary 4 of Ref. [1] applied block by block; analytic sharpness from the Fourier series of the smoothed square wave.
- **Numerics.** Wave functions on a position grid with FFT to momentum (2^{14} to 2^{15} points), Fock-space simulations with QuTiP [32], Gaussian integrals over periodic bins as theta-function sums, and energy-constrained optimization of $c_E(r)$ by semidefinite or gradient methods.

- **Data.** Binning is pure post-processing of homodyne samples [26], so existing data sets of squeezed vacuum, two-mode squeezed vacuum or cat states can be reanalyzed. A partner squeezed-light lab, or published raw data, is the experimental arm.
- **Benchmarks.** Coarse-grained entropic relations [23] and coarse-grained entanglement witnesses tested with photon pairs [24].

3.4 Numerical and experimental work packages

WP-B1. Modular variables, Zak transform and GKP background; reproduce $c(r)$ and the $r = 1/2$ checks above.

WP-B2. $c(r)$ and $\omega_{\otimes}$ for all even denominators, attainment and saturating states, and the energy-constrained overlap $c_E(r)$ at mean photon numbers 1 to 20.

WP-B3. Detector model: efficiency, electronic noise, duty-cycle and offset errors, and the survival condition of the ideal relation; deliver the tolerance map in (η, δ).

WP-B4. Applications: squeezed vacuum, the Gaussian limit L_2^{G} and its violation by cat and GKP-type states, and EPR steering with two-mode squeezed vacuum, with Rényi and Tsallis variants.

WP-B5. Analysis of real homodyne data if available; thesis and paper draft.

3.5 Expected results

- A complete answer, for binned quadratures, to open question (v) of Ref. [1]: $c(r)$ on all rationals, the role of odd denominators, and the energy-constrained overlap.
- A clean map between the qubit theory of Ref. [1] and GKP qubits: eigenstates as GKP logical states, bisector states as GKP magic states, homodyne loss as unsharpness, duty-cycle error as bias.
- A non-Gaussianity witness from binary data, with the Gaussian limit $L_2^{\mathrm{G}} \approx 1.580$ and the margins reached by realistic cat and GKP-type states.
- Steering thresholds in squeezing and efficiency, including the regime near 80% efficiency where only the corrected bound certifies steering.

3.6 Risks and mitigations

- $c(r)$ **is discontinuous and may be trivial for irrational** r**.** The energy-constrained overlap is well defined and physically relevant, and $r = 1/2$ already gives a complete story.
- **High squeezing thresholds.** The non-Gaussianity witness and the single-mode results do not need two-mode squeezing; the steering part can remain numerical.
- **No data access.** All aims except the last work package are theoretical or numerical.

4 Shared toolkit and student profile

Both projects run on one code base and the same prerequisites, so two students could work in parallel and share the majorization core.

- **Majorization core.** Lorenz curves, least concave majorants, the optimal vector from the maximal partial sums (Lemma 1 of Ref. [1]), and a library of Schur-concave measures (Shannon, Rényi, Tsallis).

- **Bound engine.** Theorem 4 in closed form, and a Corollary 4 eigenvalue solver for binary POVMs in any truncated dimension, used for the Fock space in Project A and for the Zak blocks in Project B.
- **Minimizers.** Exact entropy minima over states: the great circle for qubits, grid states and truncated Fock bases for modes.
- **Statistics.** Monte Carlo count simulation, maximum-likelihood fits and bootstrap confidence intervals.
- **Starting scripts.** The Python checks behind every number in this note are provided with it (folder `quantum-light-followups/numerics`), with a README listing what each script computes.

Table 5: The two projects side by side.

	Project A	Project B
Prerequisites	Quantum information, basic quantum optics, Python	The same, plus continuous-variable optics (Gaussian states, homodyne detection)
External resource	Photonics lab with an SPDC source and single-photon detectors, or the virtual experiment	Optional homodyne data of squeezed, two-mode squeezed or cat states
New theory	Fock-space bound for real detectors and its link to squashing	$c(r)$, energy-constrained overlap, GKP mapping, binary non-Gaussianity witness
New experiment	First tests of Propositions 1, 3, 4 and Corollary 3 with photons	Reanalysis of homodyne data with binary binning
Main risk	Systematic errors comparable to the gap Δ	Discontinuity of $c(r)$
Likely output	Thesis and an experimental paper with the partner lab	Thesis and a theory paper

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